

Crow's Foot ERD

Crow's Foot Entity-Relationship Diagram (ERD)

A **Crow's Foot Entity-Relationship Diagram (ERD)** is a another graphical method for representing the structure of a database. It shows the **entities**, their **attributes**, and the **relationships** among the entities.

Crow's Foot notation gets its name from the three-pronged symbol used to represent **many** instances in a relationship. It is widely used in database design because it clearly shows both **cardinality** and **participation** in relationships.

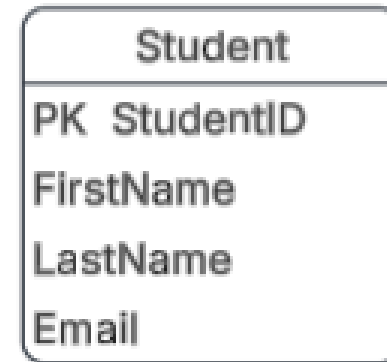
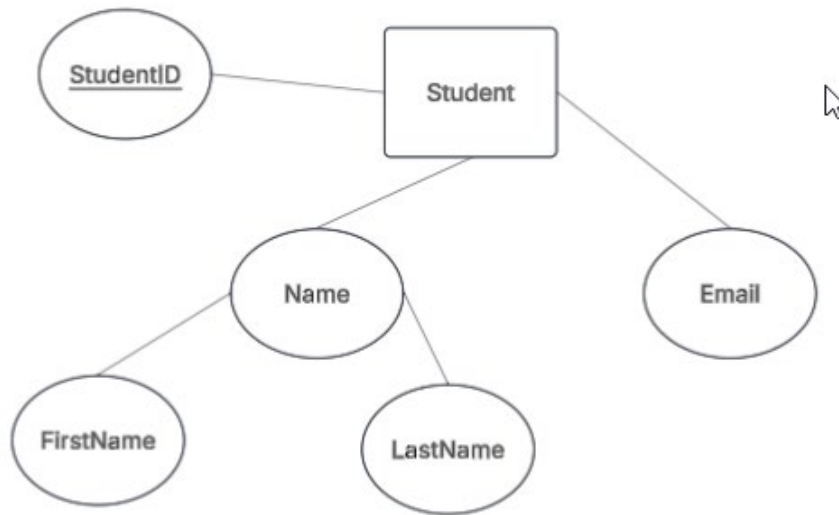


Entities

An **entity** represents a person, place, object, event, or concept about which data is stored.

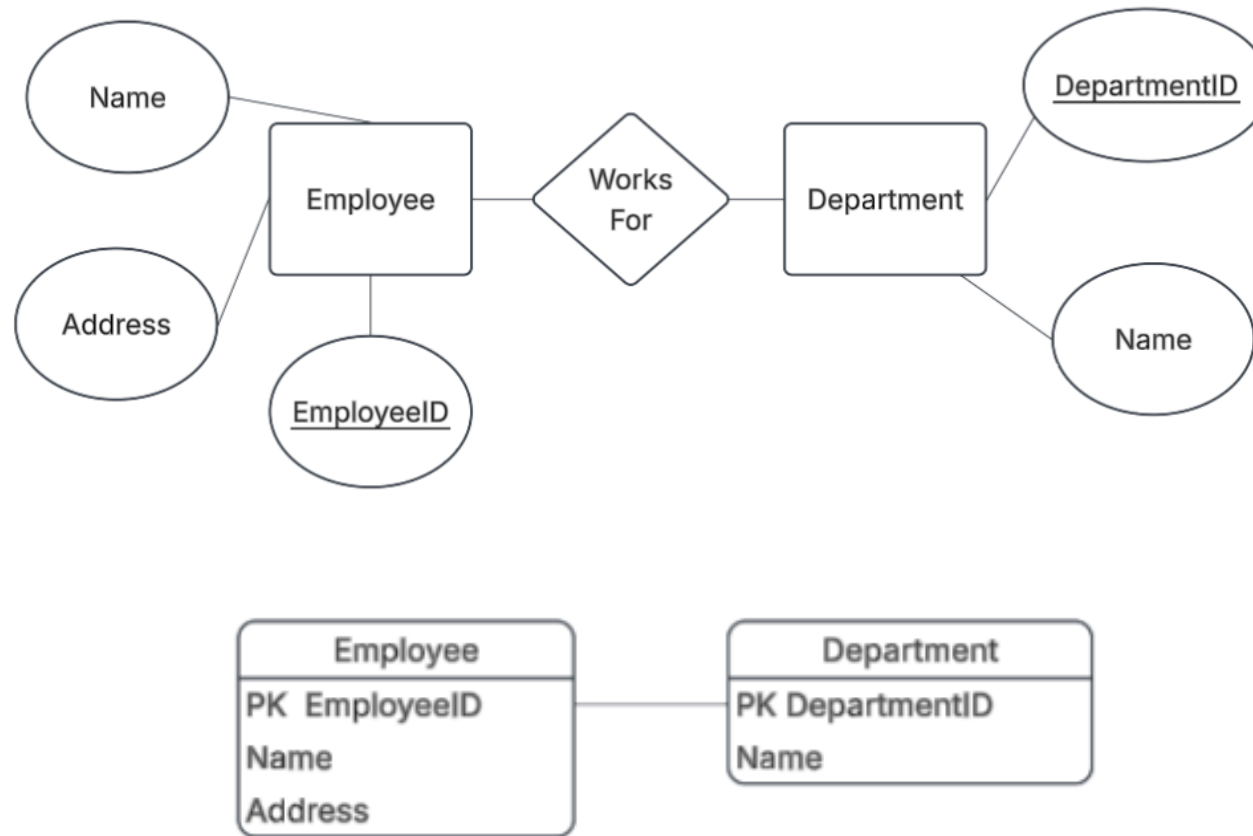
Unlike Chen notation, where an entity is usually represented by a simple rectangle and its attributes are shown as separate ovals, Crow's Foot notation normally places the attributes **inside the entity box**.

StudentID is identified as the **Primary Key (PK)**.



Relationships

The relationship is established between entities in Crow's Foot notation by drawing a straight line between the two entities.



Relationships

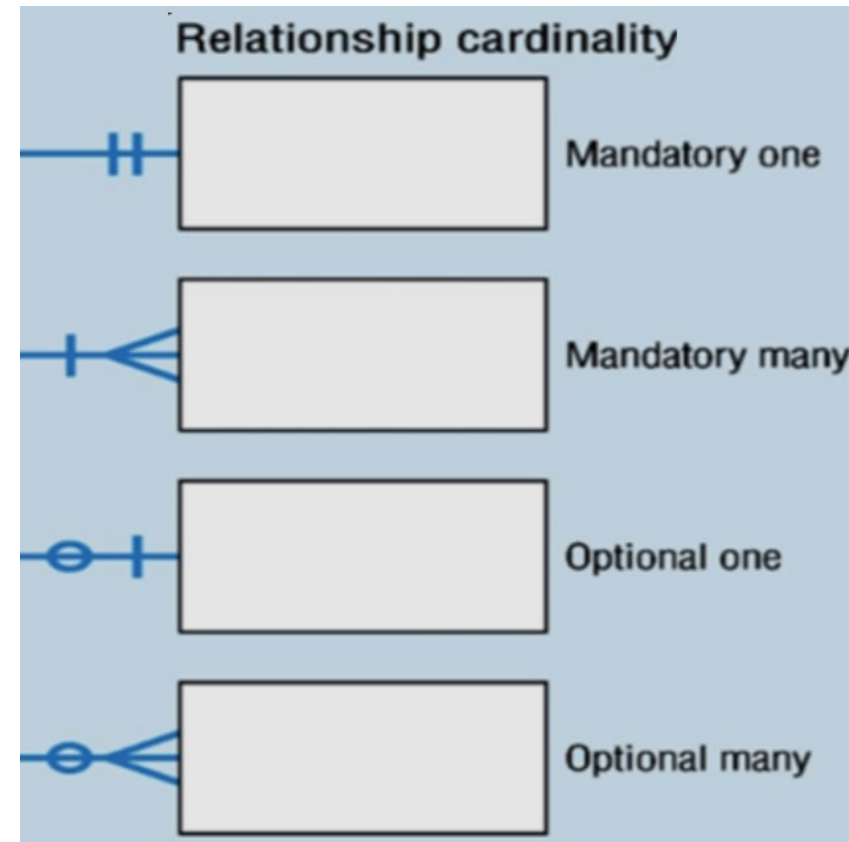
A **relationship** describes how two entities are associated with one another.

For example:

One department can have many employees.

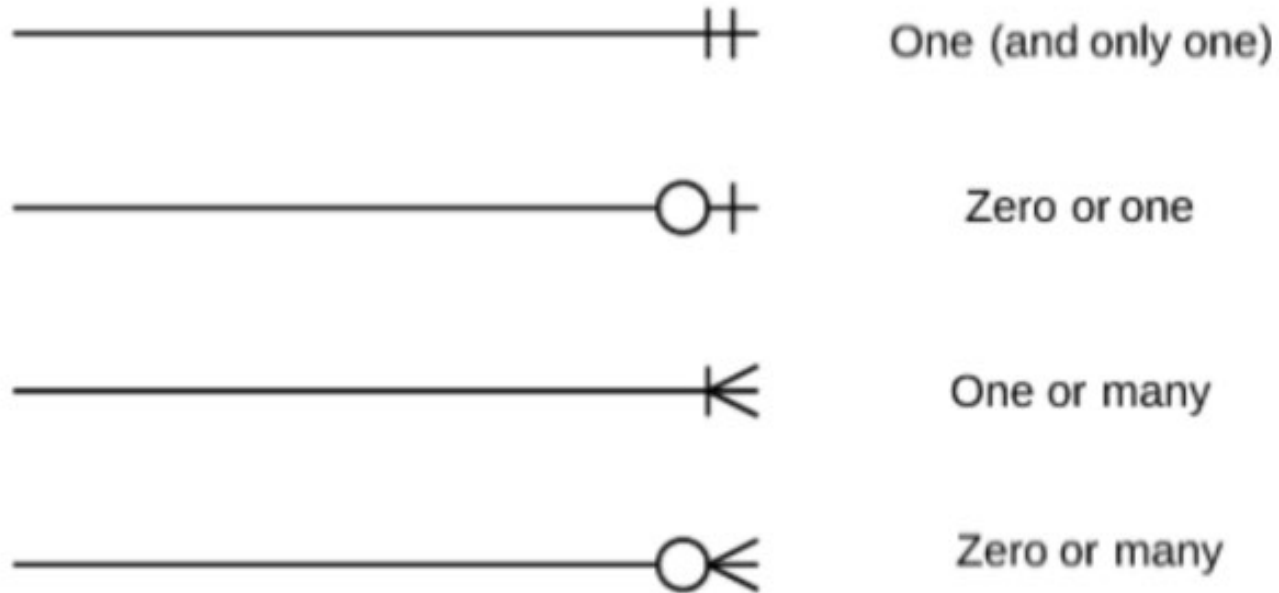
The symbols at the ends of the relationship line indicate the **minimum and maximum cardinality**.

The symbol closest to the entity indicates the **maximum cardinality**, while the symbol closer to the relationship line indicates the **minimum cardinality**.



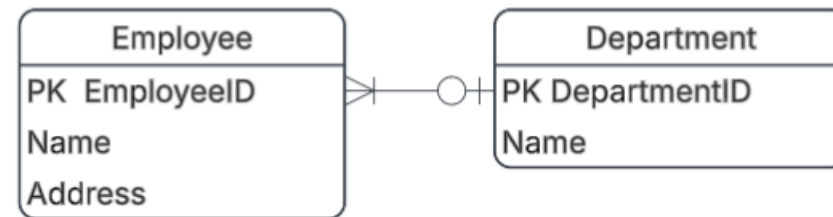
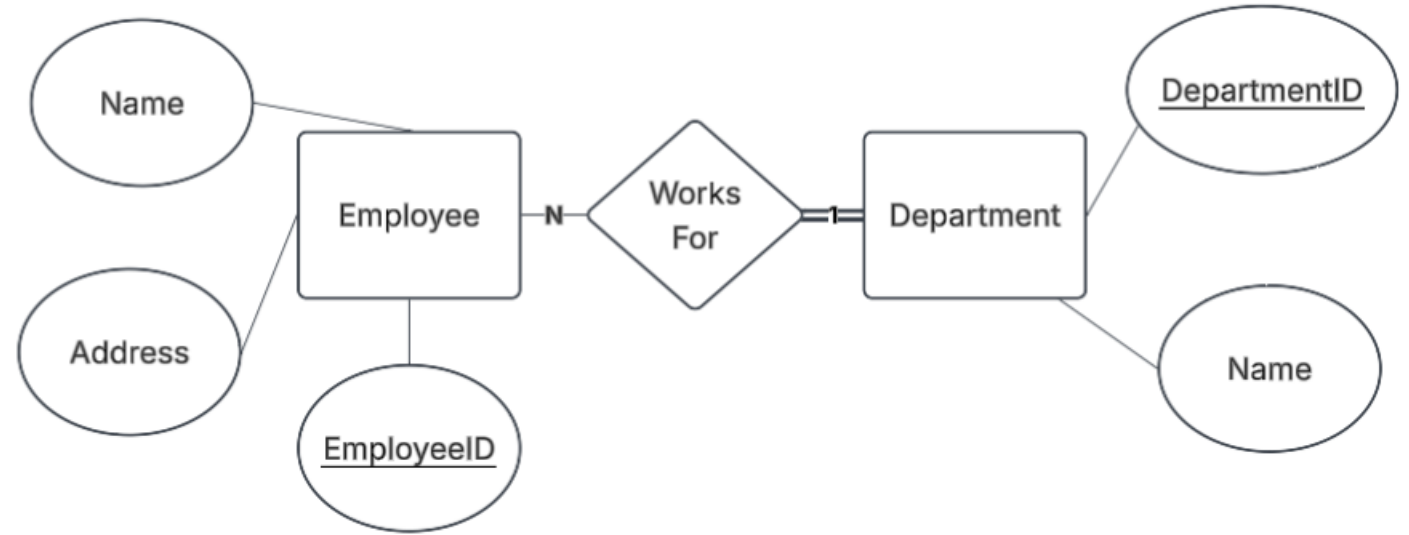
Cardinality and Ordinality

- Cardinality refers to the maximum number of times an instance in one entity can relate to instances of another entity.
- Ordinality, on the other hand, is the minimum number of times an instance in one entity can be associated with an instance in the related entity.



Going back to the Employee – Department ER

- Employees can belong to at most one department.
- Not all employees need to belong to a department.
- A department must have at least one employee and may have multiple employees.



The important rule in determining the cardinality symbol to use is is:

The symbols closest to an entity (cardinality) tell you how many instances of that entity can be associated with one instance of the entity on the opposite side.

Foreign Keys (FK) in Crow's Foot ERD

A **foreign key (FK)** is an attribute in one entity that refers to the **primary key (PK)** of another entity. It is used to **implement the relationship between the two entities** in the relational database.

For a **1:M relationship**, the general rule is:

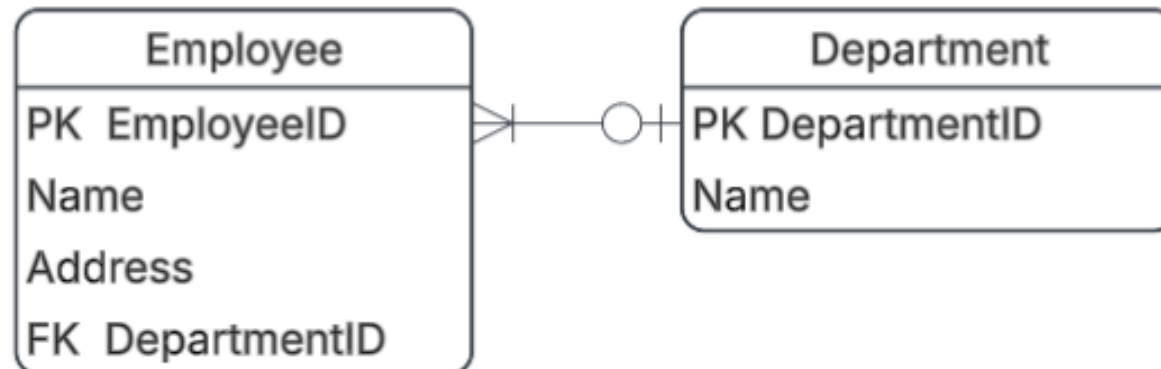
The primary key from the “1” side is placed as a foreign key in the entity on the “many” side.

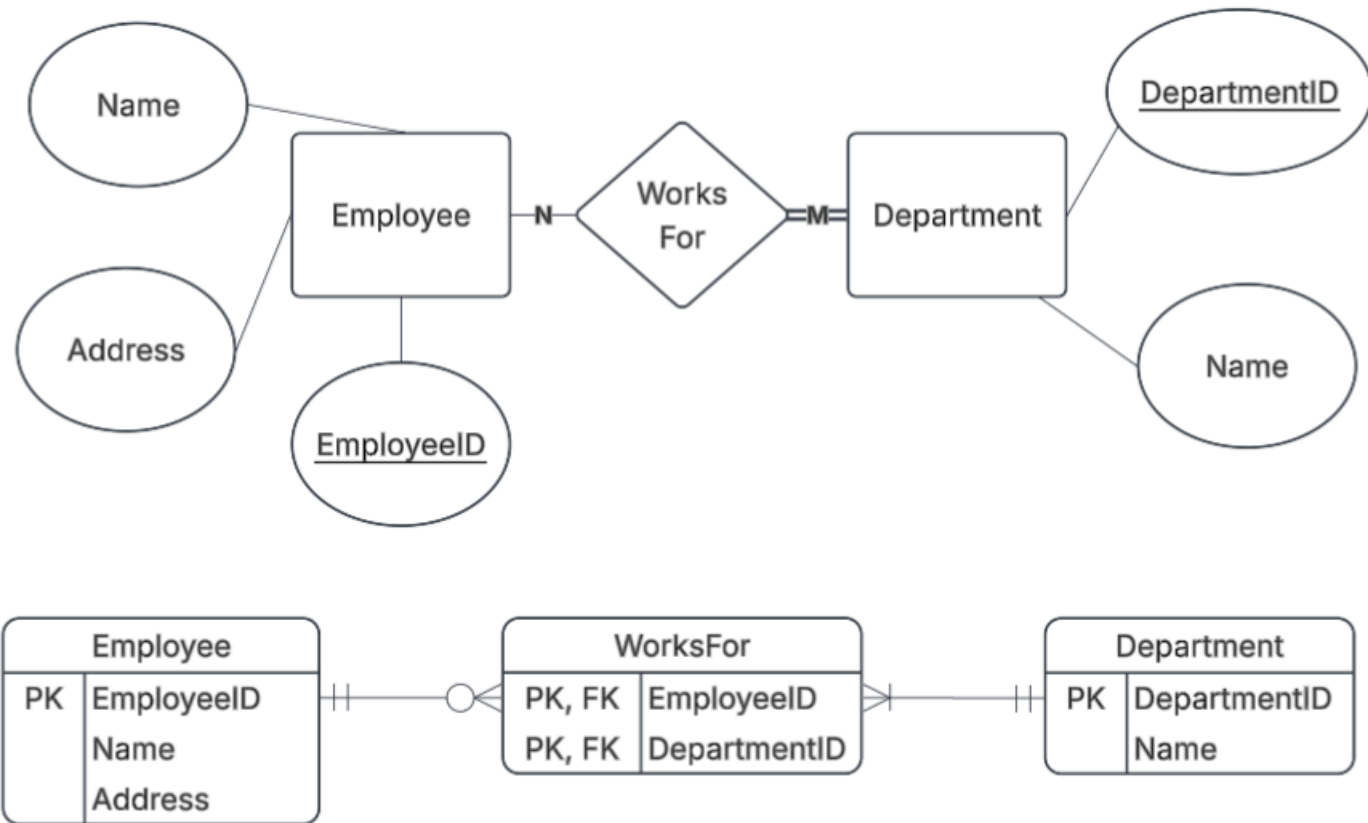
The FK placement rules are different depending on the combination of cardinalities:

1:N → FK goes on the **many side**

1:1 → FK placement depends on the business rules/optionality . Usually in a **1:1 relationship**, you normally **place the foreign key on only one side**, not both.

M:N → create an **associative entity** containing FKs from both entities





Employee — WorksFor

Near Employee: || → each WorksFor record must belong to **exactly one Employee**.

Near WorksFor: O< → one Employee may have **zero or many WorksFor records**.

Department — WorksFor

Near Department: || → each WorksFor record must belong to **exactly one Department**.

Near WorksFor: |< → one Department must have **one or many WorksFor records**.

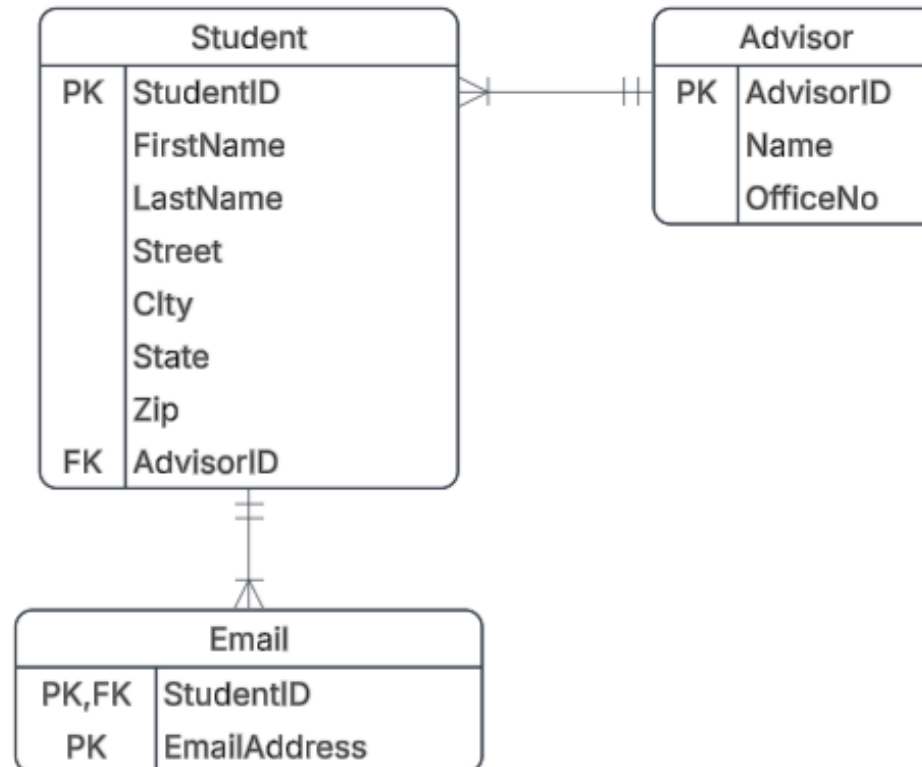
Converting an M:N Relationship to Crow's Foot Notation

When a relationship in **Chen notation** has an **M:N (many-to-many) cardinality**, an **associative entity** should be introduced when converting it to a Crow's Foot ERD.

The associative entity resolves the M:N relationship into **two 1:N relationships**.

Class Exercise 1 — Student and Academic Advisor

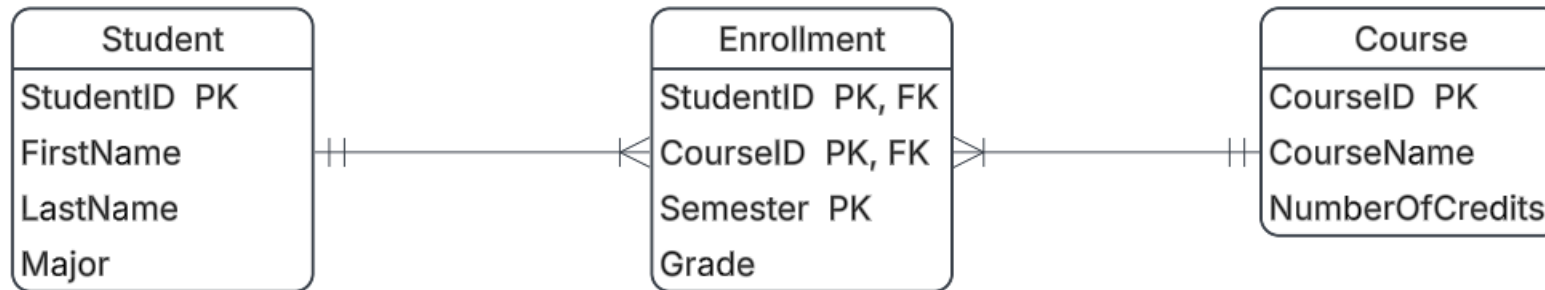
A university maintains information about its **students** and their **academic advisors**. Each student has a unique **Student ID**, a **Name** consisting of a **First Name** and **Last Name**, an **Address** consisting of **Street**, **City**, **State**, and **ZIP Code**, and one or more **Email addresses**. Each academic advisor has a unique **Advisor ID**, an **Advisor Name**, and an **Office Number**. Each student is advised by **one academic advisor**. An academic advisor advises at least one student.



Class Exercise 2 — Student, Enrollment and Course

A university maintains information about **students**, **enrollment** and **courses**.

Each student has a unique StudentID, FirstName, LastName, and Major. Each course has a unique CourseID, CourseName, and NumberOfCredits.

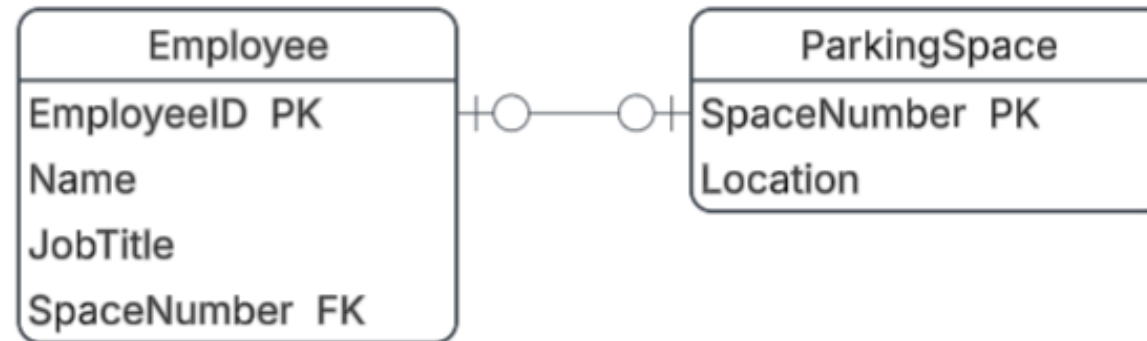


Exercise 3 — Employee and Parking Space

A company provides reserved parking spaces to some employees.

Each employee has an EmployeeID, Name, and JobTitle. Each parking space has a unique SpaceNumber and Location.

An employee may be assigned **zero or one parking space**. A parking space may be assigned to **zero or one employee**.



Superclass/Subclass in Crow's Foot Notation

In Crow's Foot notation, the **superclass and subclasses are all drawn as entity boxes**. The superclass is placed above the subclasses, and a branching line connects it to the subclass entities.

Unlike Chen notation, Crow's Foot does not have one universally standardized symbol for specialization/generalization. Depending on the software or notation variant, you may see a **branching connector**, an ISA symbol, or simply a label such as **Total/Disjoint** next to the subclass connection.

Exercise 4: University Employee Department

- A university maintains information about its **employees** and **departments**.
- Each **employee** has a unique **EmployeeID**, along with a Name and Address. Each **department** has a unique **DepartmentID** and a DepartmentName.
- An employee may belong to **zero or one department**, while each department has **one or more employees**.
- Employees are classified into two types: **Faculty** and **Staff**. Every employee must be classified as **either Faculty or Staff, but not both**.
 - For each **Faculty** member, the university records the **Rank** and **TenureStatus**.
 - For each **Staff** member, the university records the **Position**, **EmploymentType**, and **Field**.

